MILITARY SPECIFICATION

CUTTER, BARBED WIRE, INSULATED HANDLES, HAND-OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hand operated, barbed wire cutter with insulated handles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-T-570	-	Tool Steel, Alloy.
PPP-P-40	-	Packaging and Packing of Hand Tools.

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

FSC 5110

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DRAWING

ARMY NATICK LABORATORIES

2-9-42 - Cutter, Barbed Wire, Insulated Handles, Hand-Operated.

(Miniature copy of Drawing 2-9-42, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standard

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Materials and components. Materials and components shall be as specified herein and on applicable drawings. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial barbed wire cutter.

3.3 Tool steel, alloy. Alloy tool steel shall conform either to class S1 of QQ-T-570 or commercial fine grain tool steel possessing the chemical composition specified in table I (see 4.4.1.1).

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TABLE I. Tool steel chemical composition

Carbon	Silicon	Vanadium	Manganese	Phosphorus and sulfur
0.61 - 0.69	0.75 - 1.10	0.15 - 0.25	0.30 - 0.50	0.025 max.

3.4 Design and construction. The design and construction of the cutter shall be in accordance with the requirements specified herein and as shown on Drawing 2-9-42.

* 3.4.1 Jaws. The jaws shall be made from an alloy tool steel specified in 3.3.1. The bevel of the finished jaws shall have a hardness of Rockwell C53 to C61 when tested as specified in 4.6.1. The tool shall have means for adjusting cutting edges. The space between cutting edges shall be not more than 0.004 inch.

* 3.4.2 Handles and straps. The handles and straps shall be fabricated from steel in accordance with the manufacturer's standard practice. Finished handles and straps shall have a hardness of Rockwell C43 to C47 when tested as specified in 4.6.1.

* 3.4.3 Rivets. Rivets shall be fabricated from carbon steel in accordance with the manufacturer's standard practice. Before assembly, rivets shall have a hardness of Rockwell C30 to C35 when tested as specified in 4.6.1.

* 3.4.4 Handle insulation. The handle insulation shall be black and capable of being molded to a steel handle. The insulation shall have:

(1) A dielectric strength to withstand a 14,000-volt RMS, 60-Hertz, AC potential for 1 minute, with no leakage or passage of current in excess of 70 microamperes per inch of immersion per cutter handle in a saline solution when tested as specified in 4.6.2. The insulation shall show no abnormal distortion or blistering after testing in accordance with 4.6.2.

(2) Resistance to moisture absorption so that after immersion in water, the insulation shall withstand a 7,000-volt, RMS, 60-Hertz, AC potential, with no flash over (surface discharge), spark over (air discharge), or breakdown (puncture discharge), when tested as specified in 4.6.3.

(3) Resistance to impact so that the insulation of the handles shall withstand the blow of a sharp edged object and also comply with the distortion, blistering, and dielectric requirements in (1) above, when tested as specified in 4.6.4.

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(4) Resistance to oil, heat, acid, and alkali with no deterioration of the handle insulation such as blistering, swelling, peeling, becoming spongy, or being dissolved, when tested as specified in 4.6.5 through 4.6.7, as applicable.

* 3.5 Performance. When tested as specified in 4.6.8, the cutter shall make clean cuts with spring wire at all points between the innermost and outermost points of the cutting edges without: (1) Binding of movable parts; (2) Cutter parts becoming loose; (3) Indenting of chipping of the jaw edges; (4) Permanent deflection or deformation of the jaw and handles.

* 3.6 Finish application. Exposed metal surfaces of the cutter shall have a black, corrosion-resistant finish in accordance with the manufacturer's standard practice. Prior to the application of finish, the cutter surfaces shall be treated or cleaned to provide a smooth and adherent surface.

* 3.7 Marking for identification. The cutter shall be permanently marked for identification in accordance with MIL-STD-130. The marking on the assembled cutter shall be visibly located on the top jaw. The marking shall include the manufacturer's name, or readily identifiable trademark, the letters "US", and "5000 volts max.".

* 3.8 Workmanship. The completely fabricated cutter shall be free from burrs, slivers and sharp edges. Cutting edges shall be parallel, and in alignment. Parts shall be as specified, free from dents, fractures, breakage or deformation. No parts shall be damaged or impaired in any way and metal parts shall show no evidence of rust or corrosion.

* 3.8.1 Handle insulation. The handle insulation shall be smooth and thoroughly adhered to the handle steel.

* 3.8.2 Rivet joints. Heads of rivets shall be fully formed and in full contact with their respective bearing surface.

* 3.8.3 Finish. Finished surfaces shall be free from imbedded foreign matter, scratches, rust areas, and areas of no finish. Cutting edges and spun heads of rivets may have no finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required, it shall be examined as specified in 4.4.3.1 and 4.4.3.2 and subjected to tests marked "X" in column 2 of table III. The presence of any defects or failure to pass any tests shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, quality shall be based on sampling and inspection in accordance with provisions set forth in MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of reference specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.4.1.1 Material inspection (steel). The chemical composition of steel for the jaws (see 3.3.1) shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. Two samples shall be taken from each lot of material of the same form and thickness, unless the lot consists of only one piece, in which case only one sample shall be taken. Failure of any sample to comply shall be cause for rejection of the lot.

* 4.4.2 In-process inspection.

* 4.4.2.1 Process examination. Examination shall be made of the uncoated cutter to determine that metal surfaces, prior to application of the finish, have been cleaned and oxide-scale removed. Whenever any nonconformance is noted, correction shall be made to the process and all items processed.

* 4.4.2.2 Intermediate testing. Prior to assembly of the cutter, the jaws, straps, handle steel and rivets shall be tested for applicable hardness in accordance with 4.6.1. Failure to pass any hardness test shall be considered

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a defect. The inspection level shall be S-2 with an AQL (acceptable quality level) of 1.5 defects, expressed in terms of defects per hundred units. Each insulated handle, prior to the assembly of the cutter, shall be tested in accordance with 4.6.2. Failure to pass the test shall be cause for rejection of the handle.

* 4.4.3 End item inspection. The inspection lot shall be all barbed wire cutters offered for inspection at one time.

* 4.4.3.1 Visual examination. The barbed wire cutter shall be examined for defects in table II. The inspection level shall be II with an AQL of 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

* 4.4.3.2 Dimensional examination. The barbed wire shall be examined for compliance with dimension requirements. Any dimension not within the specified tolerance shall be considered a defect. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

* TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Parts missing, not as specified, not free from dents, fractures, breakage or deformation.	X	
	Not free from burrs, slivers, and sharp edges.		X
	Cutting edges not parallel and in alignment.	X	
	Handle insulation not black, smooth or thoroughly adhered to the handle steel.	X	
	Rivet heads not fully formed and in full contact with their respective bearing surfaces.	X	
	Finished surfaces not free from imbedded foreign matter, scratches, rust areas, or areas of no finish.		X

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* 4.4.3.3 End item testing. When a first article is not required, the first produced cutters shall be subject to tests marked "X" in column 3 of table III. Failure to pass any test shall be cause for rejection of the first produced items. Each sample unit shall be tested as specified in 4.6.2 through 4.6.8. Failure of any test shall be considered a defect. The inspection level shall be S-2 with an AQL of 1.5 defects, expressed in terms of defects per hundred units. The sample unit shall consist of 4 cutters. Each cutter shall be tested in accordance with test sequence in table III as follows:

(1) Cutter No. 1 - Sequence 2 through 5
(2) Cutter No. 2 - Sequence 6
(3) Cutter No. 3 - Sequence 7
(4) Cutter No. 4 - Sequence 8

* 4.4.4 Examination of preparation for delivery. Examination of preservation and packaging, packing and marking shall be in accordance with PPP-P-40. The inspection level shall be S-2.

* 4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listings, marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

* 4.6 Test methods. Tests shall be in accordance with table III.

TABLE III. Test schedule

Testing sequence	First article	Quality conformance	First produced item	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
1		X		Hardness	4.6.1	3.4.1 thru 3.4.3
2	X	X	X	Moisture absorption resistance	4.6.3	3.4.4 (2)
3	X	X	X	Impact resistance	4.6.4	3.4.4 (3)
4	X	X	X	Dielectric strength	4.6.2	3.4.4 (1)

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TABLE III. Test schedule (cont'd)

Testing sequence	First article	Quality conformance	First produced item	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
5	X	X	X	Performance	4.6.8	3.5
6	X	X	X	Oil and heat resistance	4.6.5	3.4.4 (4)
7	X	X	X	Acid resistance	4.6.6	3.4.4 (4)
8	X	X	X	Alkali resistance	4.6.7	3.4.4 (4)

* 4.6.1 Hardness test. Hardness of rivets, jaws, straps and handle steel shall be tested in accordance with ASTM E-18 to determine compliance with the hardness requirements of 3.4.1 through 3.4.3, as applicable. Hardness shall be based on the average of three readings. Hardness of jaw shall be tested on the bevel, 1/8 inch from the cutting edge. Hardness of strap rivets, and handle steel shall be tested at any locations on the surface.

* 4.6.2 Dielectric strength test. The insulated handles of the cutter shall be immersed in 1 percent salt water solution from 2 to 2-1/2 inches from the top of the insulation. The metal part of the handle shall be one electrode and the salt solution shall be the other electrode, preferably as the ground terminal. The potential between the electrodes shall be raised at a rate of 3,000 volts per second until the final test voltage of 14,000 RMS volts, 60-Hertz, AC is reached. The test voltage shall be maintained for a period of 1 minute. During this period, voltage and current readings shall be observed to determine conformance to requirements of 3.4.4 (1). The test equipment shall have a capacity of not less than 1-kilovolt ampere. Any nonconformance shall constitute failure to pass this test.

* 4.6.3 Resistance to moisture absorption test. The insulated handle shall be immersed in water for a period of 1/2 hour, then wiped dry with a cloth and immediately tested as follows. Two metal foil electrodes shall be so placed on the insulated portion of the handle that their inner edges are 1 inch apart. A potential of 7,000 RMS volts peak, 60-Hertz, AC shall be applied to the electrodes at a rate of 3,000 volts per second. The test voltage shall be applied for a period of 30 seconds to determine compliance with 3.4.4 (2). Any nonconformance shall constitute failure to pass this test.

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* 4.6.4 Resistance to impact test. The insulated handle shall be rigidly mounted. A hardened steel cutting edge, suspended as a pendulum, shall be swung to impart a blow to the handle insulation, equivalent to 5-foot pounds of energy. The cutting edge shall strike the insulation at right angles to the longitudinal axis of the handle to determine compliance with 3.4.4 (3). The cutting edge shall consist of a 90 degree wedge, 0.010 inch in width and 1-inch long. Any nonconformance shall constitute failure to pass this test.

* 4.6.5 Resistance to oil and heat test. The insulated handle shall be immersed and soaked in lubricating oil for a period of 5 hours, at 250°F., minimum, removed from the oil bath and wiped off. The insulation shall then be examined to determine compliance with the applicable requirements of 3.4.4 (4). The oil shall be SAE (Society of Automotive Engineers) No. 30. Any nonconformance shall constitute failure to pass this test.

* 4.6.6 Resistance to acid test. The insulated handle shall be immersed and soaked in a 20 percent sulfuric acid solution for a period of 1 hour, at 150 F., minimum, removed from solution and wiped off. The insulation shall then be examined to determine compliance with the applicable requirements of 3.4.4 (4). Any nonconformance shall constitute failure to pass this test.

* 4.6.7 Resistance to alkali test. The insulated handle shall be immersed and soaked in a 40 percent sodium hydroxide solution for a period of 1 hour at 150 F., minimum, removed and wiped dry. The insulation shall then be examined to determine compliance with the applicable requirements of 3.4.4 (4). Any nonconformance shall constitute failure to pass this test.

* 4.6.8 Performance test. The cutter shall make 10 complete cuts with a commercially available carbon, hard-drawn spring wire of 0.116-inch diameter. Each cut shall be made at separate points from the innermost to the outmost points of the cutting edges to determine compliance with the performance requirements of 3.5. Any nonconformance shall constitute failure to pass this test.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, and packing for level A, B, or C shall be as specified in PPP-P-40 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The cutters covered by this specification are intended for use in the field to cut barbed wire which is charged with a high voltage.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether a first article is required (see 3.1 and 6.3).
(c) Selection of applicable levels of preservation and packaging, and packing (see 5.1).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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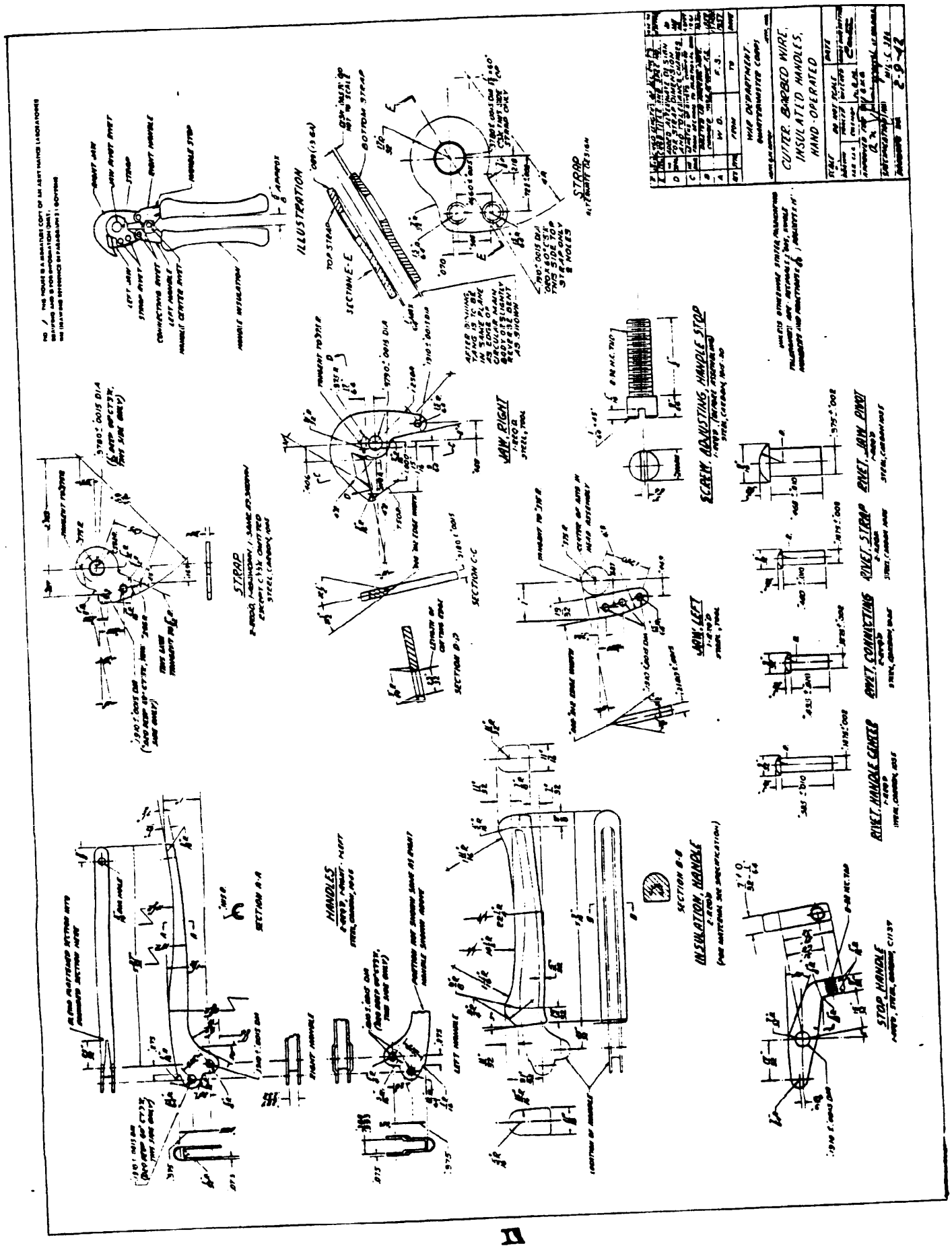
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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